Ebonite

Unless you are involved with smoking pipes, sax or clarinet mouthpieces, or a pen aficionado, you may have never heard of ebonite. In the past, it was used for a variety of things including bowling balls. Long since replaced for many applications, ebonite has faded until recently. Fountain pens and nibs were made from ebonite for many years but some companies have moved on. The renewed interest in the collectable pen market has brought a bit of a rebirth for ebonite. There are now companies producing specifically for the pen turning market. Many of the high-end custom makers tout the use of ebonite in their products. If you have an interest, you can easily work ebonite just like you would work with plastic. It cuts, sands, threads, and polishes easily. For the pen purist, ebonite is the only way to go. The modern offerings to the retail pen turning market are a joy to work with although a bit pricey.

Ebonite doesn't have to be reclaimed from old bowling balls. There are modern makers that are supplying it again

Depending on your speeds and feeds, you can slice the stuff like tape to remove stock quickly

A test fit for a custom nib blank prior to separation for finish cuts and sanding

Final dry fit prior to finish sanding and then MicroMesh to get the ebonite to pop

Solid surface materials

Last but not least, I've included solid surface materials. There are a host of these made and offered under a variety of trade names. I'll speak specifically of my experience with Corian by DuPont. Others may perform similarly but perhaps not exactly so be certain to do small-scale testing prior to committing to methods I offer here.

I've touched on Corian in the past but want to delve a bit further into it here. Corian, a filled acrylic polymer, i.e. plastic, has been around since the 1960s. Primarily a countertop and shower-surround material, it has some very admirable qualities to the turner. It is very dense and turns nicely with sharp tools. Imagine a dense wood that has no grain orientation. As such, it cuts nicely in any direction as well as takes threads without the frailty that wooden threads often present. Because there is no grain, Corian can be turned extremely thin yet retain great strength. The term solid surface was selected to tout the colour continuity throughout the material. What you see on the surface is what you can expect throughout the material.

Corian is only available in certain thicknesses. If thicker material is needed, you'll need to create it yourself by bonding multiple pieces together. Whether the same colour or a different colour, you can easily fasten Corian to itself to create blanks of your desired colours and dimension. DuPont markets the adhesive for Corian used by the professionals but it is expensive and one-time mixing. My method for fastening Corian to Corian, while not advocated by DuPont, works well for my needs. I use medium viscosity CA adhesive to bond pieces. The pieces need to be flat and clean to provide a good bond. I flatten both surfaces using a sheet of abrasive on top of a very flat surface. A surface plate, band saw, or table saw cast iron top will work well.

Once the surfaces are sanded flat as witnessed by a complete coverage of sanding dust, I clean both surfaces with denatured alcohol and refrain from touching the clean areas at all. Medium CA is spread entirely over one of the cleaned surfaces. This needs to be complete coverage since the clamping can't be relied on to spread the adhesive. Once covered, the two surfaces are put together, quickly rubbed, and immediately clamped. I use spring clamps and cover as much of the available area with clamps as possible. No accelerator is needed or is used. After resting under clamp load for at least 24 hours, the assembly is ready for use. Should a stack of more than two be required, that assembly can be then processed to be bonded

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